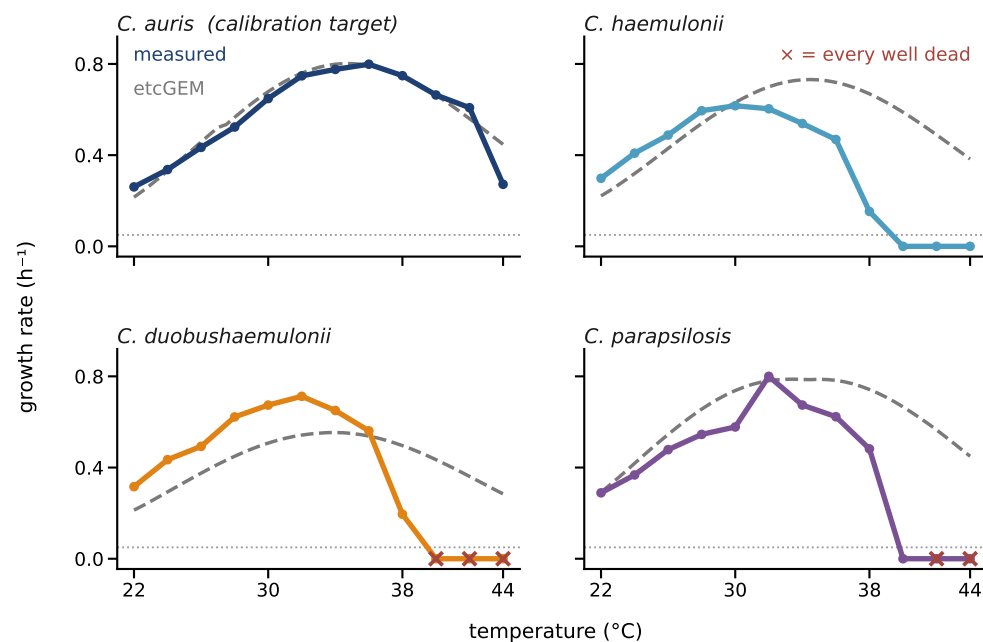
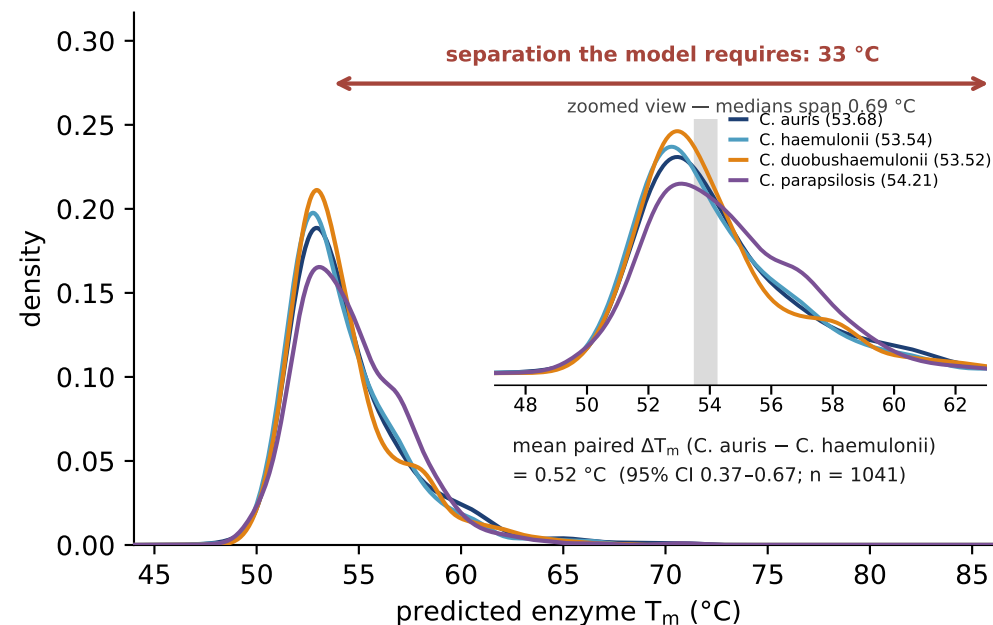


Within the etcGEM, sequence-predicted enzyme thermal properties do not reproduce the observed thermal divergence

A The model fits its calibration target but misses the relatives' collapse

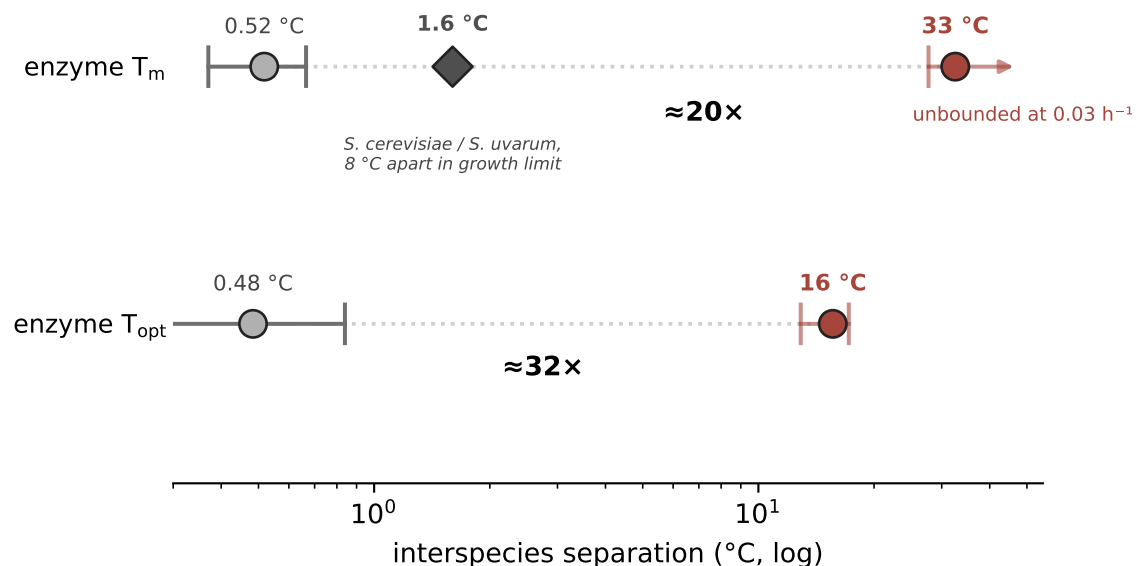


B Paired ΔT_m is directionally consistent but $\sim 63\times$ below the fitted requirement



C Required shifts exceed predictions and a measured benchmark

● sequence-predicted (paired orthologs) ● model-required ◆ measured congeneric benchmark



D Targeted perturbations do not drive relatives below detection

